

ENGR 102 Summer Bridge

Day 16 Instructor Key

Indexed Mutation and Independent Exam 2 Rehearsal

20 points • approximately 20-25 minutes

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Boolean-operator convention. Python operators appear as [and](#), [or](#), and [not](#). Their lowercase spelling is required in code.

Show intermediate state for trace credit. Program credit requires one coherent solution. Unless a prompt says otherwise, give exact Python 3 output.

Question 1. Trace indexed mutation and post-mutation state.

4 points

```
data = [2, 11, -3, 8]
edits = 0
for index in range(len(data)):
    if data[index] < 0:
        data[index] = 0
        edits = edits + 1
    elif data[index] > 10:
        data[index] = 10
        edits = edits + 1
print(data, edits)
```

Give list state after every pass and exact output.

Answer and explanation

Index 0 leaves [2, 11, -3, 8], edits 0. Index 1 caps 11, giving [2, 10, -3, 8], edits 1. Index 2 replaces -3, giving [2, 10, 0, 8], edits 2. Index 3 is unchanged. Exact output: [2, 10, 0, 8] 2.

Question 2. Repair non-mutating code.

2 points

```
data = [-2, 5]
for value in data:
    if value < 0:
        value = 0
print(data)
```

Explain why the list is unchanged and give the essential loop repair.

Answer and explanation

value receives each stored value, but assigning value = 0 only rebinds that loop variable; it does not assign to a list position. Use for index in range(len(data)) and assign data[index] = 0 when data[index] < 0.

Question 3. Program construction**14 points**

Write one complete Python program that satisfies every requirement.

- Use measurements = [9, -1, 12, 21, 6].
- Mutate negative values to 0 and values above 15 to 15.
- Count modified positions and compute the total from final values.
- Print final list, modified count, and total on separate lines.
- Do not use sum or a comprehension. Give one test in which both the first and last positions require changes.

Answer and explanation

```
measurements = [9, -1, 12, 21, 6]
modified = 0
total = 0

for index in range(len(measurements)):
    if measurements[index] < 0:
        measurements[index] = 0
        modified = modified + 1
    elif measurements[index] > 15:
        measurements[index] = 15
        modified = modified + 1
    total = total + measurements[index]

print(measurements)
print(modified)
print(total)
```

The final list is [9, 0, 12, 15, 6], modified is 2, and total is 42. The total update follows mutation so it uses final values. Boundary test [-2, 7, 20] -> [0, 7, 15], modified 2, total 22. This protects both the first and last index.

Scoring guidance

3 indexed traversal, 4 two-sided mutation, 2 exact modified count, 2 post-mutation total, 1 output, 2 first/last boundary test with expected result.